P5 Electricity

Physics | Classes 6, 7 & 8 Combined

Classes 6 • 7 • 8 | TET Paper II Science

A · ELECTRIC CHARGE — Basics & Transfer

Electric Charge	Basic property of matter that causes objects to attract or repel — carried by protons and electrons
SI Unit of Charge	Coulomb (C) — denoted by letter q
Elementary Charge (e)	Smallest charge that can exist freely = 1.602×10^{-19} C — charge on one proton or one electron
Positive Charge	Carried by PROTONS — less electrons than protons
Negative Charge	Carried by ELECTRONS — more electrons than protons
Neutral Atom	Equal number of protons and electrons — net charge = ZERO
Like charges	REPEL each other (+ and + repel | – and – repel)
Unlike charges	ATTRACT each other (+ and – attract)
Law of Conservation	Charges can NEITHER be created NOR destroyed — total charge remains constant

Transfer of Charges — Three Methods

Method	How It Works	Example
Transfer by Friction	When two materials are rubbed together, electrons transfer from one to other — both get charged	Comb rubbed on dry hair → gains electrons → attracts paper bits; Glass rod + silk cloth; Ebonite rod + fur
Transfer by Conduction	Direct contact of charged body with uncharged body → charges flow from charged to uncharged	Charged ebonite rod touches paper cylinder → cylinder gets charged and repels rod
Transfer by Induction	Charged body brought NEAR (NOT touching) uncharged body → opposite charge appears at near end	Negatively charged rod near neutral rod → near end becomes positive, far end negative — no contact needed

■ Critical Distinction — Friction Charging

■ A neutral object becomes positively charged by LOSING electrons — NOT by gaining positive charges

■ Glass rod rubbed with silk: glass LOSES electrons → glass becomes +ve; silk GAINS electrons → silk becomes –ve

■ Ebonite rod rubbed with fur: fur LOSES electrons → fur becomes +ve; ebonite GAINS electrons → ebonite becomes –ve

■ Wet hair reduces friction → fewer electrons transfer → comb attracts less paper

B · ELECTROSCOPE — Detection of Charge

Electroscope	Scientific instrument to DETECT the presence and type of electric charge on a body
Working Principle	Like charges repel — when charge is given, gold leaves move apart
Inventor	William Gilbert (British physician) — invented first electroscope in 1600
Gold-leaf type	Developed by Abraham Bennet (British scientist) in 1787 — gold/silver used (best conductors)

Gold-Leaf Electroscope — Structure & Working

- Structure: Glass jar + vertical brass rod through cork + brass disc at top + two gold leaves hanging from rod inside jar
- When charged body touches brass disc → charge transfers to gold leaves → leaves have same charge → REPEL → leaves DIVERGE (spread apart)
- More charge → greater divergence of gold leaves
- When charged body brought NEAR (not touching) → opposite charge appears at leaves → leaves still diverge
- Gold and silver used because they are the BEST conductors of electricity
- Pith-ball electroscope: simpler type — small light pith ball hung by silk thread; attracted/repelled by charges
- Electroscope detects PRESENCE of charge; cannot always determine the TYPE (positive or negative)

C · CURRENT, POTENTIAL DIFFERENCE & RESISTANCE

Electric Current (I)	Flow of electric charges through a conductor — measured in Ampere (A) — symbol: I
Conventional Current	Flow of POSITIVE charges from + to – terminal (positive to negative) — direction: high to low potential
Electron Flow	Flow of ELECTRONS from – to + terminal (negative to positive) — opposite to conventional current

Potential Difference (V)	Energy needed to move 1 unit of charge between two points — SI unit: Volt (V)
Resistance (R)	Opposition to flow of electric charges — SI unit: Ohm (Ω) — R = V/I

Electric Current	**$I = q / t$** (q = charge (coulombs), t = time (seconds))

Resistance	**$R = V / I$** (V = voltage (volts), I = current (amperes))

Units of Electric Current — Conversions

1 Ampere (A)	= flow of 1 coulomb per second through a cross-section — 1 A = 1 C/s
1 milliampere (mA)	= 10^{-3} A = 1/1000 A (for small currents)
1 microampere (μA)	= 10^{-6} A = 1/10,00,000 A (for very small currents)
1 coulomb (C)	≈ charge of 6.242×10^{18} protons or electrons — 1 C = 1 A × 1 s

Conventional Current vs Electron Flow

	Conventional Current	Electron Flow
Direction	Positive terminal (+) to Negative terminal (–) of battery	Negative terminal (–) to Positive terminal (+) of battery
Charge carrier	Imaginary positive charges (historical assumption)	Actual electrons (negatively charged particles)
When established?	Before discovery of electrons (historical)	After discovery of electrons by J.J. Thomson (1897)
Usage today	Still used in circuit diagrams and calculations	Actual physical reality of current in conductors

Electrical Conductivity & Resistivity

Conductivity (σ)	Measure of material's ability to conduct electricity — SI unit: Siemens/metre (S/m) — symbol: σ (sigma)
Resistivity (ρ)	Fundamental property opposing current flow — SI unit: ohm-metre (Ω·m) — symbol: ρ (rho)

- Higher conductivity = lower resistivity → better conductor (metals have low resistivity)
- Silver has highest conductivity of all metals → best conductor
- Copper: resistivity 1.68×10^{-8} Ω·m | Silver: 1.59×10^{-8} Ω·m | Aluminium: 2.82×10^{-8} Ω·m

Measuring Instruments

Quantity	Instrument	How Connected	Unit
Current (I)	Ammeter	IN SERIES with circuit component	Ampere (A)
Potential Difference (V)	Voltmeter	IN PARALLEL across the component	Volt (V)
Resistance (R)	Ohmmeter	When circuit is OFF (no current flowing)	Ohm (Ω)

D · ELECTRIC CIRCUITS — Series & Parallel

Electric Circuit	Complete closed path through which electric current flows — electrons travel from – to + terminal
Open Circuit	Circuit with a gap/break — current DOES NOT flow
Closed Circuit	Complete path with no breaks — current flows freely
Short Circuit	Low-resistance connection between two conductors — causes excessive current, heat, sparks

Circuit Components & Symbols

Cell	Single unit of electrical energy — longer line = positive (+) terminal, shorter line = negative (–) terminal in symbol
Battery	Two or more cells connected together — provides potential difference to drive current
Switch	Device to open or close a circuit — controls current flow (ON = closed circuit; OFF = open circuit)
Bulb (Lamp)	Electrical resistor that converts electrical energy to light and heat
Fuse	Safety device — thin wire that MELTS when current exceeds safe limit → breaks circuit
Ammeter	Measures current — connected in series (low resistance)
Voltmeter	Measures voltage — connected in parallel (high resistance)

Series vs Parallel Circuits — Complete Comparison

Feature	Series Circuit	Parallel Circuit
Connection	Components connected END to END in a single loop	Components connected between SAME two points — multiple paths (branches)

Feature	Series Circuit	Parallel Circuit
Current (I)	SAME current flows through ALL components $I = I_1 = I_2 = I_3$	Current DIVIDES across components $I = I_1 + I_2 + I_3$
Voltage (V)	Voltage DIVIDES across components $V = V_1 + V_2 + V_3$	SAME voltage across all components $V = V_1 = V_2 = V_3$
Total Resistance	Total $R = R_1 + R_2 + R_3$ (increases)	$1/R = 1/R_1 + 1/R_2 + 1/R_3$ (decreases — less than smallest resistor)
Brightness of bulbs	Dimmer — each bulb shares power from battery	Brighter — each bulb gets full voltage
If one bulb fails	ALL bulbs go OFF — circuit breaks	OTHER bulbs stay ON — other paths still complete
Everyday use	Rarely — series lights in older decoration strings	HOME WIRING — all appliances/rooms; if one fails, others still work

■ **Homes use PARALLEL circuits — independent switches for each room, each appliance works independently at full voltage.**

E · CONDUCTORS & INSULATORS

Type	Definition	Why?	Examples
Conductor	Material that ALLOWS electric current to pass easily — low resistance	Has many FREE ELECTRONS in outermost orbit — loosely bound and can move freely through material	Metals: Silver (best), Copper, Aluminium, Iron, Gold; also Salt water, human body
Insulator (Bad Conductor)	Material that DOES NOT allow current to pass easily — very high resistance	Electrons in outermost orbit are tightly bound — cannot move freely — very few free electrons	Rubber, Plastic, Wood, Glass, Ceramic, Dry air, Pure water, Silk, Cotton, Paper
Semiconductor	Partially conducts — between conductor and insulator in properties	Moderate number of free electrons — conductivity changes with temperature and impurities	Silicon (Si), Germanium (Ge) — used in transistors, diodes, solar cells, computer chips

- Rubber gloves used by electricians — rubber is an insulator → protects against electric shock
- Electric wires coated with plastic/rubber insulation — prevents accidental shock
- Human body conducts electricity — touching live wire is dangerous (body acts as conductor to earth)
- Dry air = insulator; Moist/humid air = partial conductor (lightning possible)

- Distilled/pure water = insulator; Salt water (with dissolved ions) = conductor

F · LIGHTNING, EARTHING & LIGHTNING ARRESTER

Lightning	Massive discharge of static electricity between charged clouds or between cloud and Earth — enormous heat and light
Earthing (Grounding)	Safety measure connecting exposed metal parts of electrical devices to Earth via low-resistance wire — prevents shock
Lightning Arrester	Device protecting buildings from lightning — metallic rod (spikes) at top of building connected to ground by copper cable

Three-Wire System in Domestic Supply

Live wire (L)	Carries current AT HIGH VOLTAGE from power station to appliance — colour: RED (old) or BROWN (new)
Neutral wire (N)	Returns current back to power station — stays near zero voltage — colour: BLACK (old) or BLUE (new)
Earth wire (E)	Connected to metallic body of appliance and to earth — safety wire — colour: GREEN or YELLOW-GREEN

- Earth wire function: if live wire touches metal body (insulation failure) → current flows to earth safely → prevents shock
- Car during lightning storm: SAFE — car body acts as Faraday cage, protects occupants from electric field
- During lightning: DO NOT stand in open ground — squat down (make yourself small); avoid tall trees

G · EFFECTS OF ELECTRIC CURRENT — Three Main Effects

■ **Three effects of electric current: (1) Heating Effect (2) Magnetic Effect (3) Chemical Effect**

1. Heating Effect of Electric Current

Definition	When current passes through a conductor, electrical energy converts to HEAT energy — due to friction between electrons and conductor molecules
Factors	Heat produced depends on: (1) Amount of current (2) Resistance of wire (3) Time for which current flows
Nichrome	Alloy of Nickel + Iron + Chromium — used as heating element — HIGH melting point + HIGH resistance
Tungsten	Used in bulb filament — HIGH melting point (3,422°C) + HIGH resistance → glows bright white when heated

Fuse wire	Made of Lead + Tin alloy — LOW melting point → melts quickly when excess current flows → breaks circuit → protects appliances
MCB	Miniature Circuit Breaker — replaces fuse — breaks circuit automatically when excess current; can be switched back ON without replacing wire

Applications of Heating Effect

Appliance	How Heating Effect Is Used
Electric Bulb	Tungsten filament heated to ~2500°C → emits visible white light
Electric Iron	Nichrome heating element heats metal plate → heat conducted to clothes
Geyser / Water Heater	Immersion heater coil heats water — convection distributes heat throughout water
Electric Kettle	Heating element at bottom heats water — convection carries heat upward
Electric Cooker	Coils turn red hot → heat conducted to cooking pot
Arc Welding	Practical application of short circuit — enormous heat generated at arc point melts and fuses metals

2. Magnetic Effect of Electric Current

Discovery	Hans Christian Oersted (Danish physicist) discovered in 1819 that electric current creates magnetic field around conductor
Electromagnet	Coil of insulated wire wound around iron nail/core — becomes magnet ONLY when current flows — loses magnetism when current off
Making electromagnet	Wind insulated wire tightly around iron nail in form of coil; connect to battery — nail becomes magnet
Key property	Polarity of electromagnet CHANGES when direction of current changes

Applications of Magnetic Effect

Application	How Magnetic Effect Is Used
Electric Bell	Current through electromagnet → attracts hammer → hammer strikes bell → spring pulls hammer back → circuit breaks → repeats
Electric Crane	Powerful electromagnet lifts heavy iron/steel loads; current switched off to drop the load
Telephone	Changing magnetic field causes diaphragm (thin metal sheet) to vibrate → produces sound

Application	How Magnetic Effect Is Used
Electric Motor	Current-carrying coil in magnetic field → rotates → converts electrical to mechanical energy
Hospital (eye surgery)	Electromagnets remove iron/steel splinters from eye without surgery

3. Chemical Effect of Electric Current

Definition	When current passes through a conducting liquid (electrolyte), chemical reactions occur — compounds decompose into ions
Electrolysis	Decomposition of molecules of a solution into positive and negative ions by passing electric current through it
Electrolyte	Conducting liquid/solution that allows current to pass and undergoes chemical change — e.g., copper sulphate solution, salt water, acids
Anode (+)	Positive electrode — metal plate connected to positive terminal of battery
Cathode (–)	Negative electrode — object to be plated — connected to negative terminal of battery

Electroplating — Most Common Application

Electroplating	Depositing a thin layer of one metal over the surface of another metal by passing electric current
Why?	Protect cheaper metals from corrosion, rust; give attractive appearance; reduce cost vs solid precious metal objects
Anode	Pure metal plate (metal to be deposited) — e.g., copper plate; gradually dissolves
Cathode	Object to be plated — e.g., iron spoon; metal deposits here
Electrolyte	Salt solution of metal to be deposited — e.g., copper sulphate for copper plating

Electroplating Applications

Object / Purpose	Metal Deposited	Why That Metal?
Iron bridges, automobiles (anti-rust)	Zinc (galvanisation)	Zinc does not corrode; cheap; protects iron from rusting
Car parts, bath taps, bicycle handles, kitchen gas burners	Chromium	Shiny appearance; does not corrode; resists scratches; hard

Object / Purpose	Metal Deposited	Why That Metal?
Jewellery (artificial gold/silver)	Gold or Silver	Attractive appearance; tarnish-resistant; cheaper than solid gold/silver
Food cans (tin cans)	Tin	Non-toxic; does not corrode; safe for food contact
Circuit boards (electronics)	Copper or Gold	Excellent conductor; prevents oxidation; reliable contacts

H · Solved Numerical Problems

Problem 1	If 30 coulombs of electric charge flows through a wire in 2 minutes, find the current.
Given	$q = 30$ C; $t = 2$ min $= 2 \times 60 = 120$ s
Solution	$I = q/t = 30/120 = 0.25$ A
Answer	**Current = 0.25 A**

Problem 2	0.002 A current flows through a circuit. Convert this to microamperes (μA).
Given	$I = 0.002$ A; 1 A $= 10^6$ μA
Solution	0.002 A $= 0.002 \times 10^6$ μA $= 2 \times 10^3$ μA
Answer	**2000 μA (2×10^3 μA)**

Problem 3	The voltage across a resistor is 12 V and the current through it is 4 A. Find resistance.
Given	$V = 12$ V; $I = 4$ A
Solution	$R = V/I = 12/4 = 3\ \Omega$
Answer	**Resistance = 3 Ω (3 ohms)**

Problem 4	Two resistors R■ = 4 Ω and R■ = 6 Ω connected in SERIES. Find total resistance.
Given	$R_1 = 4\ \Omega$; $R_2 = 6\ \Omega$; Series connection
Solution	R_total $= R_1 + R_2 = 4 + 6 = 10\ \Omega$
Answer	**Total resistance = 10 Ω**

Problem 5	Two resistors R■ = 4 Ω and R■ = 6 Ω connected in PARALLEL. Find total resistance.
Given	$R_1 = 4\ \Omega$; $R_2 = 6\ \Omega$; Parallel connection
Solution	$1/R = 1/R_1 + 1/R_2 = 1/4 + 1/6 = 3/12 + 2/12 = 5/12 \rightarrow R = 12/5 = 2.4\ \Omega$

Answer	**Total resistance = 2.4 Ω (less than smallest resistor = 4 Ω)**

Problem 6	A charge of 240 C passes through a conductor in 4 minutes. Find current in mA.
Given	q = 240 C; t = 4 min = 240 s
Solution	$I = q/t = 240/240 = 1\ A = 1 \times 10^{3}\ mA$
Answer	**Current = 1 A = 1000 mA**

I · Key Facts — Rapid Revision (Numbers, Years & Names only)

Formulas, Values & Units

Quantity / Concept	Formula / Value	Unit / Note
Electric Current (I)	I = q/t (charge/time)	Ampere (A)
1 Ampere	= 1 coulomb per second (1 C/s)	Definition
1 milliampere	$= 10^{-3}$ A (0.001 A)	Conversion
1 microampere	$= 10^{-6}$ A (0.000001 A)	Conversion
Resistance (R)	R = V/I (Voltage/Current)	Ohm (Ω)
Resistance — Series	$R = R_1 + R_2 + R_3$ (total increases)	Same current
Resistance — Parallel	$1/R = 1/R_1 + 1/R_2 + 1/R_3$ (total decreases)	Same voltage
Voltage — Series	$V = V_1 + V_2 + V_3$ (divides)	Volt (V)
Current — Parallel	$I = I_1 + I_2 + I_3$ (divides)	Ampere (A)
Elementary charge (e)	$= 1.602 \times 10^{-19}$ C	Proton or electron charge
1 coulomb (C)	$\approx 6.242 \times 10^{18}$ elementary charges	1 A × 1 s
Conductivity unit	Siemens/metre (S/m)	σ (sigma)
Resistivity unit	Ohm-metre (Ω·m)	ρ (rho)
Silver resistivity	1.59×10^{-8} Ω·m (lowest)	Best conductor
Copper resistivity	1.68×10^{-8} Ω·m	Common conductor

Scientists, Discoveries & Years

Scientist / Event	Contribution / Fact	Year
William Gilbert	British physician — invented FIRST electroscope (first electrical instrument)	1600
Hans Christian Oersted	Danish physicist — discovered MAGNETIC EFFECT of electric current	1819
Abraham Bennet	British scientist — developed gold-leaf electroscope	1787
Thomas Alva Edison	American inventor — lit 14,000 bulbs in 9,000 New York houses (first power grid)	1882
Calcutta Electric Supply	First thermal power plant in India — Calcutta (Kolkata)	1899 (Apr 17)
J.J. Thomson	Discovered electron — proved electron flow in conductors	1897
Basin Bridge Madras	First power station in Madras city — powered government press, hospital, tramways	~1900
Nichrome discovery	Alloy of Ni+Fe+Cr — used in heating elements — high melting point	Early 1900s

J · Glossary — Definitions

Electric Charge — Basic property of matter that causes attraction or repulsion; carried by protons and electrons; unit: Coulomb	**Elementary Charge (e)** — Smallest freely existing charge = 1.602×10^{-19} C; charge on one proton or one electron
Coulomb (C) — SI unit of electric charge; 1 C = charge of ≈ 6.242×10^{18} protons/electrons	**Electric Current (I)** — Flow of electric charges through a conductor per unit time; $I = q/t$; SI unit: Ampere (A)
Ampere (A) — SI unit of electric current; 1 A = 1 coulomb per second	**Conventional Current** — Flow of positive charges from + to – terminal; historical concept; opposite to electron flow
Electron Flow — Actual flow of electrons from – to + terminal; actual physical current	**Potential Difference (V)** — Energy needed to move 1 unit of charge between two points; causes current to flow; SI unit: Volt (V)
Resistance (R) — Opposition to flow of electric current; $R = V/I$; SI unit: Ohm (Ω)	**Conductivity (σ)** — Measure of material's ability to conduct electricity; SI unit: Siemens per metre (S/m)
Resistivity (ρ) — Property of material opposing current flow; SI unit: Ohm-metre (Ω·m)	**Conductor** — Material with many free electrons that allows current to pass easily — metals, salt water
Insulator — Material with tightly bound electrons that resists current flow — rubber, plastic, wood, glass	**Semiconductor** — Material between conductor and insulator; conductivity changes with conditions — Silicon, Germanium

Ammeter — Instrument to measure electric current; connected in SERIES; unit: Ampere	**Voltmeter** — Instrument to measure potential difference; connected in PARALLEL; unit: Volt
Ohmmeter — Instrument to measure resistance; used when circuit is OFF; unit: Ohm	**Electric Circuit** — Complete closed path through which electric current flows
Series Circuit — Components connected end to end in single loop; same current; voltage divides	**Parallel Circuit** — Components connected between same two points; voltage same; current divides
Fuse — Safety device made of lead-tin alloy with low melting point; melts on excess current; breaks circuit	**MCB (Miniature Circuit Breaker)** — Safety device replacing fuse; breaks circuit automatically; can be switched back on without replacing
Short Circuit — Low-resistance connection between conductors; causes excess current, heat, sparks, damage	**Earthing** — Connecting exposed metal parts of electrical appliances to Earth via low-resistance wire; prevents shock
Lightning — Massive electrostatic discharge between charged clouds or cloud and Earth	**Lightning Arrester** — Metallic rod on building connected to ground via copper cable; safely directs lightning to Earth
Electroscope — Instrument to detect presence of electric charge; gold leaves repel when charged	**Induction** — Charging an uncharged body by bringing a charged body near it without physical contact
Heating Effect — Electric current through conductor converts electrical energy to heat; basis of electric bulb, heater, iron	**Magnetic Effect** — Electric current creates magnetic field around conductor; discovered by Oersted (1819)
Electromagnet — Coil of wire wound around iron core; becomes magnet only when current flows	**Chemical Effect** — Electric current passing through electrolyte causes chemical changes; basis of electrolysis
Electrolysis — Decomposition of liquid compound into ions by passing electric current through it	**Electroplating** — Depositing a layer of one metal on another using electrolysis; protects and decorates metals
Electrolyte — Conducting liquid that carries current and undergoes chemical change in electrolysis	**Nichrome** — Alloy of Nickel, Iron and Chromium; high melting point and resistance; used in heating elements
Tungsten — Metal with highest melting point (3422°C); used in bulb filaments; glows white when very hot	